



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

PHYSICS

PAPER 1 Multiple Choice

4023/1

1 hour

SPECIMEN PAPER

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended.)

Electronic calculator

INSTRUCTIONS TO CANDIDATES

Answer **all** questions. For each question, there are **four** possible answers, **A, B, C** and **D**.

Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

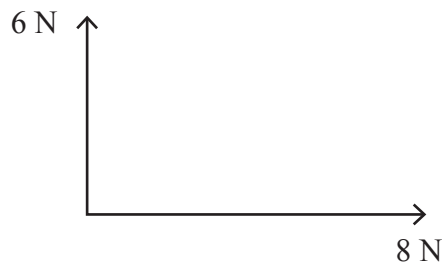
This specimen paper consists of 11 printed pages.

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1 Which one is the SI unit of temperature?

- A $^{\circ}\text{C}$
- B $^{\circ}\text{K}$
- C K
- D C

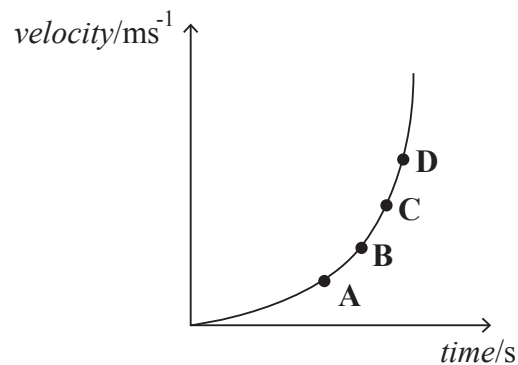
2 The diagram shows two forces that are perpendicular to each other.



What is the magnitude of the resultant force?

- A 2 N
- B 10 N
- C 14 N
- D 48 N

3 The diagram shows a *velocity-time* graph of an athlete competing in a race.



At which point is the acceleration greatest?

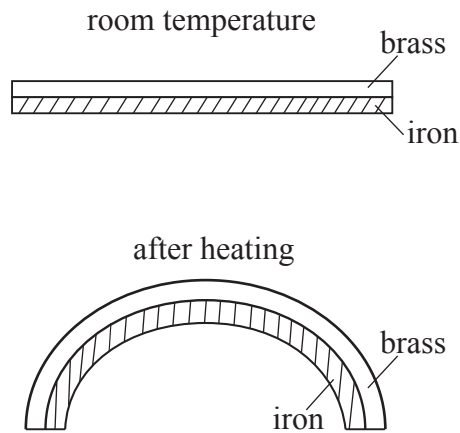
4 Terminal velocity of an object is the velocity

- A of a falling object when the acceleration is zero.
- B of a falling object when acceleration is increasing.
- C when the upward force is greater than the downward force.
- D when the downward force is greater than the upward force.

- 5 A parachutist demonstrating some skills at the national independence celebrations jumps off a helicopter. The acceleration of the parachutist
- A is decreasing uniformly.
 - B is increasing uniformly.
 - C remains constant.
 - D is decreasing non-uniformly.
- 6 How do air pressure patterns help to predict weather?
- A High pressure always indicate rain.
 - B Low pressure indicates clear skies.
 - C Changes in pressure can indicate approaching weather systems.
 - D Pressure patterns have no relation to weather.
- 7 It is better to use a long spanner rather than a short one to tighten a bolt because
- A a longer spanner reduces the friction between the bolt and the nut.
 - B a longer spanner produces more torque with the same force.
 - C a longer spanner increases the pressure applied directly on the bolt head.
 - D a longer spanner makes the bolt thread stronger.
- 8 A dancer of mass 60 kg jumps and lands on the floor with a velocity of 5 m/s.
- What is the dancer's momentum just before landing?
- A 300 kgms^{-1}
 - B 65 kgms^{-1}
 - C 12 kgms^{-1}
 - D 0.08 kgms^{-1}
- 9 When an ox-drawn cart accelerates up a hill
- A it gains both kinetic energy and potential energy.
 - B it gains kinetic energy but loses potential energy.
 - C it gains potential energy but loses kinetic energy.
 - D both kinetic energy and potential energy decrease.
- 10 A bucket of maize seed of mass 20 000 g is raised through a vertical height of 10 m.
- What is the work done?
- A 20 J
 - B 200 J
 - C 2 000 J
 - D 20 000 J

- 11 Machines are lubricated in industry in order to
- A increase energy gained.
 - B increase efficiency.
 - C increase velocity ratio.
 - D reduce mechanical advantage.
- 12 The mechanical advantage of a single moveable pulley is
- A 1.
 - B 2.
 - C 3.
 - D 4.
- 13 Which joining method is most appropriate to join plastic to plastic materials?
- A gluing
 - B nailing
 - C brazing
 - D soldering
- 14 What shape is most commonly used in trusses for stability?
- A square
 - B circle
 - C rectangle
 - D triangle

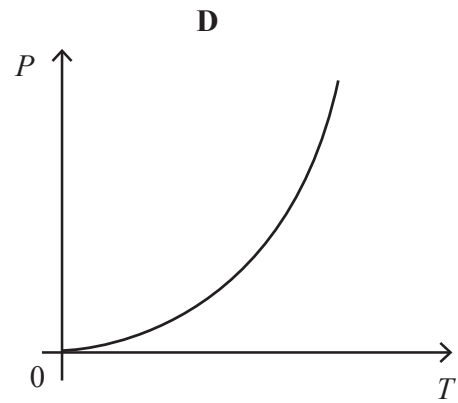
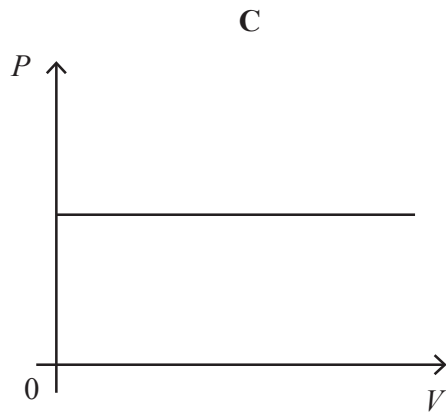
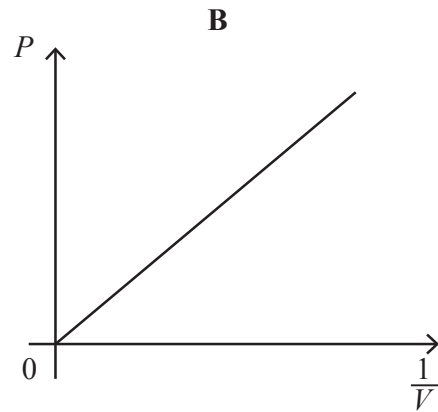
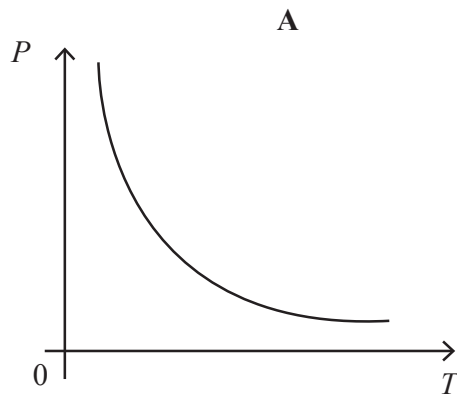
- 15** The diagrams show a bimetallic strip when it is at room temperature and after it has been heated during use.



The change in shape occurs because

- A** brass expands more than iron.
 - B** iron expands more than brass.
 - C** both expand at the same rate.
 - D** neither brass nor iron expands.
- 16** In a vacuum flask both walls are silvered in order to
- A** reduce heat gain by conduction only.
 - B** reduce heat gain or heat loss by radiation.
 - C** reduce heat gain or heat loss by convection.
 - D** reduce heat gain or heat loss by conduction and convection.

17 Which graph illustrates Boyle's Law?



18 Modern car manufacturers prefer engines with multiple cylinders.

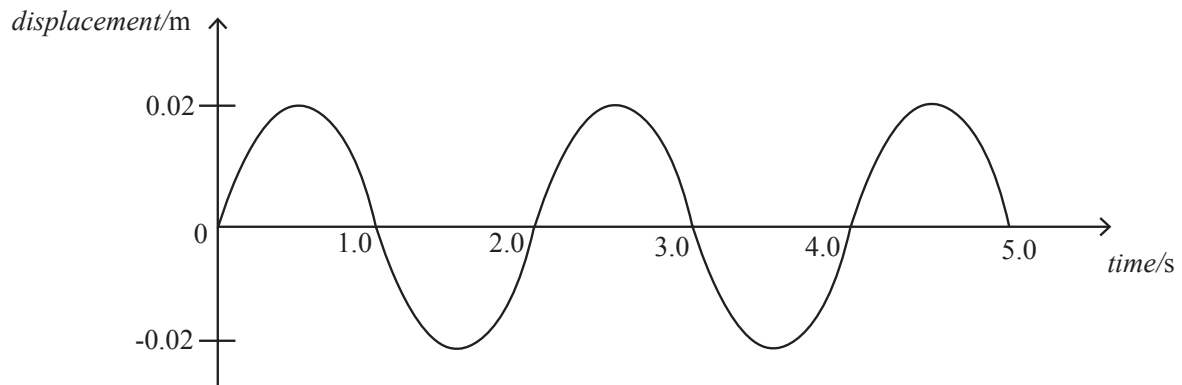
Which statement is **not** true about the use of multiple cylinders?

- A** They have more power strokes per revolution.
- B** They increase the car's acceleration.
- C** They facilitate the function of computers in the car.
- D** They reduce the vibration of the car.

19 The efficiency of a car engine can be measured by the

- A** car's speed.
- B** sound produced.
- C** smoke produced.
- D** fuel economy.

- 20 The diagram shows how *displacement* varies with *time* for a wave passing through a point.



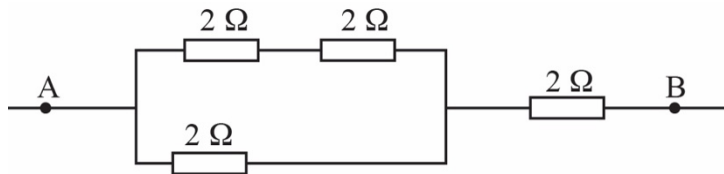
What is the frequency of the wave?

- A 0.25 Hz
 - B 0.50 Hz
 - C 1.00 Hz
 - D 2.00 Hz
- 21 A sound wave is emitted downwards from a ship to a seabed a distance, d .
- Which expression can be used to find the speed of sound in seawater if the time taken for the echo to be detected at the ship is t ?
- A $\frac{d}{t}$
 - B $d \times t$
 - C $\frac{t}{d}$
 - D $\frac{2d}{t}$
- 22 A student observes that a swimming pool appears shallower than its actual depth.
- Which phenomenon is responsible for this effect?
- A dispersion of light
 - B refraction of light
 - C reflection of light
 - D convergence of light

23 The correct formula used to calculate electrical power is

- A $P = VI.$
- B $P = \frac{V}{I}.$
- C $P = VI t.$
- D $P = I^2 R t.$

24 The diagram shows a combination of resistors.



What is the effective resistance between A and B?

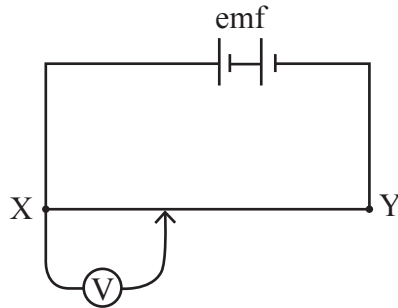
- A 1.33 Ω
 - B 2.00 Ω
 - C 3.33 Ω
 - D 8.00 Ω
- 25 How long would it take for a charge of 10 C to flow through a cellphone, when there is a current of 2 A?
- A 20.0 s
 - B 12.0 s
 - C 5.0 s
 - D 0.2 s
- 26 What is the resistance of a lamp when the voltage across it is 12 V and the current flowing through it is 4 A?
- A 48.0 Ω
 - B 16.0 Ω
 - C 3.0 Ω
 - D 0.3 Ω
- 27 In mining communities, magnets were sometimes used to separate iron-rich ores from the soil.

Which material is most strongly attracted to a magnet?

- A copper
- B aluminium
- C iron
- D wood

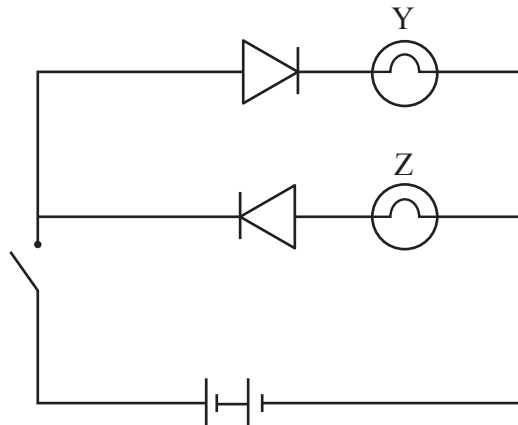
- 28 In a relay switch, the electromagnet is used to
- A store electrical energy.
 - B reduce resistance in the circuit.
 - C convert mechanical energy into heat.
 - D pull a contact to close or open a circuit.
- 29 The voltage generated by an alternating current generator cannot be increased by
- A turning the coil more rapidly.
 - B using a coil with a smaller surface area.
 - C using a coil with more turns.
 - D increasing the strength of the magnet.
- 30 In rural Zimbabwe, electricity from power stations must be transmitted over long distances to reach villages.
- Transformers are used in the process to
- A change A.C to D.C.
 - B store electricity for later use.
 - C increase the brightness of bulbs.
 - D reduce energy losses by stepping up voltage.
- 31 When a student builds a simple d.c electric motor, what is the role of the commutator?
- A to increase the magnetic field strength
 - B to reverse the direction of current flow
 - C to reduce friction between the moving parts
 - D to insulate the coil from the surroundings
- 32 Which form of energy is possessed by an electron emitted from a hot metal surface by thermionic emission?
- A nuclear energy
 - B chemical energy
 - C kinetic energy
 - D gravitational potential energy

- 33 The diagram shows a potentiometer with a resistive wire XY.



What happens to the voltmeter reading when the sliding contact is moved towards Y?

- A remains constant
 - B becomes zero
 - C increases
 - D decreases
- 34 The diagram shows an experiment set up by a Physics learner.



What happens to bulbs Y and Z when the switch is closed?

	bulb Y	bulb Z
A	on	off
B	off	off
C	on	on
D	off	on

- 35** What is an actuator?
- A** A device which converts energy to motion.
 - B** A device which converts a.c to d.c.
 - C** A device which converts electrical energy to heat.
 - D** A device which converts electrical energy to sound.
- 36** The following are disadvantages of using robots except that robots
- A** lead to high unemployment.
 - B** lead to deskilling.
 - C** work the whole day without stopping.
 - D** are expensive to buy.
- 37** What is the primary purpose of a robot's control system?
- A** to provide power to the robot
 - B** to sense the environment
 - C** to execute programmed instructions
 - D** to communicate with humans
- 38** How many electrons are there in a neutral atom of sodium $^{23}_{11}\text{Na}$?
- A** 11
 - B** 12
 - C** 23
 - D** 34
- 39** Which one is an isotope of Uranium ($^{235}_{92}\text{U}$)?
- A** $^{92}_{235}\text{U}$
 - B** $^{92}_{238}\text{U}$
 - C** $^{238}_{146}\text{U}$
 - D** $^{238}_{92}\text{U}$
- 40** A miner buys 100 g of a radioactive substance with a half-life of 2 years which decays to a stable compound.
- What mass of radioactive substance remains after 6 years?
- A** 0 g
 - B** 12.5 g
 - C** 25.0 g
 - D** 87.5 g